Understanding the Line-vs-Point Test: A Variant of the Low Degree Test

Pushpendra Pal

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1 Introduction

The Line-vs-Point Test is a simplified version of the Low Degree Test (LDT) used in zero-knowledge proofs and probabilistically checkable proofs (PCPs). It helps verify that a function $s : \mathbb{F}^n \to \mathbb{F}$ behaves like a low-degree polynomial without requiring its full specification.

2 The Line-vs-Point Test Procedure

Goal

Check whether a function s is close to a polynomial of degree $\leq d$.

Steps

1. **Pick a random line:** Choose a random affine line $\ell(t) = a + t \cdot b$ where $a, b \in \mathbb{F}^n$.
2. **Evaluate s on the line:** Query s at $d + 1$ distinct values $t_0, ..., t_d$.
3. **Interpolate:** Use Lagrange interpolation to find the unique degree-d polynomial $p(t)$ that fits the evaluations.
4. **Check:** Optionally evaluate s at a new t' on the same line and check if $s(\ell(t')) = p(t')$.

Why This Works

If s is close to a degree-d polynomial, it will consistently agree with low-degree interpolants along random lines. If s is not, the test will detect inconsistencies with high probability.

3 Clarifications and Misconceptions

- The interpolated polynomial need not be constant—it just must be of degree $\leq d$.
- LDT does not merely interpolate but tests whether s behaves like a low-degree polynomial globally.
- Even high-degree or random functions may pass the test on some lines by coincidence, hence the use of randomness in repeated testing.

4 Examples

Example 1: Function That Passes the Test

Let $\mathbb{F}_5 = \{0, 1, 2, 3, 4\}$ and define:

$$s(x, y) = 3x + 2y \mod 5$$

This is a degree-1 polynomial (multilinear).

Choose a line: $\ell(t) = (1 + t, 2 + t)$. Then:

$$s(1, 2) = 2, \quad s(2, 3) = 2, \quad s(3, 4) = 2$$

These values lie on the constant polynomial $p(t) = 2$, which is of degree $0 \leq 1$. The test **passes**.

Example 2: Function That Sometimes Fools the Test

Let:

$$s(x, y) = x^2 + y^2 \mod 5$$

This is degree 2 in each variable, not multilinear.

Line: $\ell(t) = (1 + t, 1 + 2t)$

$$s(1, 1) = 2, \quad s(2, 3) = 3, \quad s(3, 0) = 4$$

These fit the degree-1 polynomial $p(t) = 2 + t$, so the test **passes**—but falsely.

Example 3: Failing Case Revealing True Degree

Try $\ell(t) = (t, t)$:

$$s(t, t) = t^2 + t^2 = 2t^2$$

This is degree 2 in t, so the test **fails**.

5 Constructing the Polynomial

Given evaluations $s(\ell(t_0)), ..., s(\ell(t_d))$, interpolate the degree-d polynomial:

$$p(t) = \sum_{j=0}^{d} y_j \cdot \ell_j(t) \quad \text{where} \quad \ell_j(t) = \prod_{\substack{0 \leq i \leq d \\ i \neq j}} \frac{t - t_i}{t_j - t_i}$$

Then verify consistency with additional queries.

6 Summary

- The Line-vs-Point Test checks if a function s agrees with a low-degree polynomial on random lines.
- Passing on all lines implies closeness to a degree-d polynomial.
- Failing on some lines reveals that s is not globally low-degree.
- It is widely used in zero-knowledge proof systems like zk-SNARKs and STARKs.